

Responses to the Reviewers' Comments on the manuscript with ID TCOM-TPS-25-2414.R1

Model-Based Beam-Steered Optical Wireless Positioning with Single-LED Single-Photodiode for 3D Localization

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We would like to thank the Editor and the Reviewers for their careful reading of the revised manuscript and for their constructive comments, which have helped us further improve the accuracy of the claims and the presentation quality of the paper. We hope that the present version addresses all remaining comments and suggestions.

In this document, we provide detailed explanations of the changes made in the revised version of our article to address these comments. Each comment is reminded in **red** and is followed by our response, in black font. The changes made to the revised manuscript are mentioned in **blue**. Please also note that, unless stated otherwise, all page, section, figure, table, equation, and reference numbers, as well as acronyms, refer to the revised manuscript. In the marked-up version of the revised manuscript, added text is shown in blue and removed text in red.

Responses to the Editor

The revised manuscript has been improved, and the main technical concerns raised in the previous round have largely been addressed. In particular, the proposed two-stage procedure and the distinction between direction finding and distance recovery are now much clearer. One important consistency issue, however, remains. The direction-finding stage operates without receiver reorientation, whereas completion of the 3D position requires one cooperative PD reorientation for distance recovery. This distinction is correctly explained in several parts of the revised manuscript. However, statements such as “full 3D indoor localization without receiver rotation” still appear, particularly in the Conclusion. Such wording overstates the capability of the complete positioning procedure and may affect the credibility of the claimed practical advantage. This point should therefore be corrected consistently throughout the manuscript. The authors should also carefully address the remaining reviewer comments (see also the attached file) and clearly acknowledge the simulation-based nature and practical limitations of the present validation.

We would like to thank the Editor for the positive assessment of our revised manuscript and for this clear summary of the remaining points, which we address in detail in the following sections. We fully agree with the Editor's remark on the remaining consistency issue. **The main change made in this revision is to distinguish, with a single formulation used throughout the manuscript, the direction-finding stage—which operates with the receiver held at an arbitrary, unknown orientation—from the completion of the 3D position, which requires a single cooperative PD reorientation for distance recovery.** The corresponding modifications, together with the explicit acknowledgment of the simulation-based nature and practical limitations of the validation, are summarized here:

- **Conclusion (Section VIII):** “without receiver rotation” has been removed from the first paragraph, which now states the requirement of each stage (page 13); in the third paragraph, the simulations “support” the framework “within the scope of the simulation-based validation delimited in Section VII-D”, and experimental validation on a hardware prototype is listed as the first future work item (page 14).
- **Abstract (page 1):** the cooperative measurement is described as one “in which the PD is reoriented once toward the LED”, and the tilt robustness is attributed to the direction-finding stage.
- **Section I:** the paragraph preceding the contributions (page 2) and the first contribution (page 3) now mention the single cooperative PD reorientation of the distance-recovery step.
- **Table I (page 2):** the entry of the proposed system is tagged with explicit “(D-F)” and “(3D)” labels, and its footnote has been rewritten.
- **Section VII-A-2 (page 10):** the robustness to receiver tilt is stated to be specific to the direction-finding stage.
- **New Section VII-D, “Limitations of the Present Validation” (page 13):** acknowledges the simulation-based nature of the results and the absence of experimental validation, states the system requirements presupposed by the validation (including the actuated receiver needed for the cooperative alignment), and delimits the receiver-orientation independence under NLOS multipath.
- **Responses to Reviewer 3:** physical explanation of the WLS trend (Section VII-B), definition of the 14 dB operating point (Section IV; Fig. 5 re-plotted on the same SNR scale as Fig. 10), initialization of the genetic algorithm (Section IV), applications in edge computing and digital twins (Section I), and explicit tilt/azimuth labels in Table II.
- **Editorial corrections:** the GA selection operator is now correctly described as stochastic uniform selection (Section IV), the acronyms IRS and WKNN are defined, and an equation cross-reference in Section III-A has been corrected.

Responses to Reviewer 1

No further comments.

We would like to thank the reviewer for the time devoted to the evaluation of our manuscript and for the positive assessment of the revised version.

Responses to Reviewer 2

The revised manuscript has addressed the major logical concern and is now much clearer and more coherent. The remaining issues are related primarily to practical validation and presentation clarity rather than fundamental technical flaws.

We sincerely thank the reviewer for the positive assessment of the revised manuscript and for the remaining comments, which we fully share and have addressed as detailed below.

1. *The direction-finding stage can operate with arbitrary receiver orientation, whereas the complete 3D localization pipeline still requires cooperative PD reorientation for distance recovery. This should be clarified. The manuscript should still ensure that the Abstract,*

Table I, and contribution statements do not overstate the receiver-orientation-free nature of the complete 3D positioning system.

We agree. A single formulation is now used throughout the manuscript: the direction-finding stage operates under an arbitrary (unknown, fixed) receiver orientation, whereas completing the 3D position requires a single cooperative PD reorientation for distance recovery. The corresponding changes are the following.

Abstract (page 1):

“[...] to first estimate the LED-to-PD direction, and then recover the distance from a single cooperative beam-aligned measurement, in which the PD is reoriented once toward the LED.”

Section I, paragraph preceding the contributions (page 2):

“[...] the direction-finding stage requires zero receiver movement, and the entire beam-steering complexity resides in the infrastructure; on the receiver side, a single cooperative PD reorientation is required only at the final distance-recovery step.”

First contribution (Section I, page 3):

“[...] are mathematically independent of the receiver orientation $\mathbf{n}_r$, enabling direction finding without receiver pose knowledge, followed by a single cooperative PD reorientation for distance recovery.”

Conclusion (Section VIII, first paragraph, page 13):

“This paper introduced a beam-steered single-LED, single-PD OWP architecture that delivers full 3D indoor localization without cameras or PD arrays: the direction-finding stage operates with the receiver held at an arbitrary, unknown orientation, while completing the 3D position requires a single cooperative PD reorientation for distance recovery.”

Table I (page 2) has been relabeled as detailed in our response to Comment 3, and the new Section VII-D (page 13) lists the actuated receiver required for the cooperative alignment as a system requirement of the complete pipeline that does not apply to direction finding (see our response to Comment 2).

- 2. There is one practical limitation remaining: the evaluation is still mainly simulation-based, so the absence of experimental validation or a more realistic hardware-implemented study should be more explicitly acknowledged.*

We agree. A new subsection, Section VII-D “Limitations of the Present Validation” (page 13), has been added at the end of the results section. It reads:

“The results reported in this paper are obtained from Monte Carlo simulations of the LOS channel model (1)–(3) under AWGN; no experimental validation is included at this stage. Beyond the modeling assumptions of Section II, the validation presupposes two system requirements: (i) accurate steering control, such that the commanded orientations $\{\mathbf{n}_{t,i}\}$ are reproduced without pointing error; and (ii) for the distance-recovery step, an actuated receiver coordinated with the transmitter to execute the cooperative alignment—a requirement of the complete pipeline that does not apply to direction finding.

The LOS channel model also delimits the receiver-orientation independence of Section VI-C: under NLOS multipath, the received powers include reflected contributions that depend on both the LED steering orientation and $\mathbf{n}_r$ and cancel in neither (29) nor (24), so the property holds only approximately and is not guaranteed when reflections dominate; in indoor channels with an unobstructed LOS, however, the LOS component dominates the reflected contributions [44].

The reported accuracies should therefore be read as the performance attainable under these conditions; their experimental assessment on a prototype of the proposed architecture is part of our ongoing work (Section VIII)."

Consistently, the third paragraph of the Conclusion (page 14) now states that the simulations "support the effectiveness of the proposed framework, within the scope of the simulation-based validation delimited in Section VII-D" (instead of "confirmed"), and future work item (i) (page 14) reads "experimental validation of the complete two-stage pipeline on a hardware prototype (cf. Section VII-D)". The experimental validation of the complete pipeline is part of our ongoing work.

3. *In Table I, the notation used for the proposed method, especially the "Arbitrary / Controlled / D-F" description of receiver orientation, is somewhat difficult to parse.*

We thank the reviewer for this remark. In the revised Table I (page 2), each line of the "Ours" row is now tagged with the stage it refers to, and the footnote has been rewritten:

- **Rx Orientation:** "Arbitrary (D-F)" / "One reorientation (3D)" (the former "Controlled" has been removed);
 - **Error (APE):** "0.54° (D-F)" / "2.00 cm (3D)";
 - **Footnote:** "†Two-stage system: during direction finding (D-F), the PD is held at any fixed, unknown orientation (arbitrary), without any receiver rotation; completing the 3D position requires a single cooperative PD reorientation toward the transmitter (one reorientation). See Section VI-C."
4. *The statements about robustness to random receiver tilts should clearly indicate that this property mainly applies to the direction-finding stage rather than to the complete distance-recovery measurement.*

We agree. The tilt-robustness evaluation of Section VII-A-2 concerns the direction-finding stage only, since the distance-recovery measurement is performed under cooperative alignment. The scope is now explicit in the two statements where it was not:

- Abstract (page 1): "[...] with direction-finding robustness to random receiver tilts (degradation below 3%)."
- Section VII-A-2 (page 10): "This robustness is specific to the direction-finding stage, as distance recovery is performed under cooperative alignment (Section II)."

Responses to Reviewer 3

We sincerely thank the reviewer for the careful reading of the revised manuscript and for the constructive comments, which have helped us clarify several technical points. Our responses follow.

1. *The reviewer recommends providing a physical or geometric explanation of why the WLS performance degrades when moving from $K=5$ to $K=9$ orientations in Table VI, as it seems counterintuitive compared to the GLS and NLS trends.*

We thank the reviewer for this question. All $K - 1$ power ratios $\hat{\beta}_i = (\hat{\mu}_i/\hat{\mu}_1)^{1/m}$ share the same reference measurement $\hat{\mu}_1$. From (35), the error of each ratio contains an independent term n_i/μ_i and a common term $-n_1/\mu_1$, identical for all i ; combining (37)–(38), the ratio-error covariance has the exact diagonal-plus-rank-one form $\Sigma_\beta \propto \text{diag}(\beta_i^2/\mu_i^2) + \mu_1^{-2}\beta\beta^\top$, whose rank-one term is the noise of the shared reference. Geometrically, since $\hat{\mathbf{a}}_i = \mathbf{a}_i - \tilde{n}_i\mathbf{n}_{t,1}$ (Section VI-A), this common term displaces all constraint normals coherently along $\mathbf{n}_{t,1}$ and rotates the estimated direction systematically; unlike independent errors, it does not average out when orientations are added. GLS accounts for this structure exactly through Σ_β^{-1} , which attenuates the common component by $1/(1 + \sum_{i \geq 2} \mu_i^2/\mu_1^2)$, a factor that strengthens with K . WLS retains only the diagonal and treats the single coherent perturbation as $K - 1$ independent ones; this mismatch grows with the number of ratios sharing the reference (4 at $K = 5$, 8 at $K = 9$), which is also why Section VI-B recommends GLS when the cross-correlations are not negligible. We further note that the penalty is confined to the RMSE: in Table VI, the APE (2.25 $\rightarrow$ 2.15 cm) and the $\text{CDF}_{90\%}$ (4.49 $\rightarrow$ 4.01 cm) of WLS improve from $K = 5$ to $K = 9$, so the RMSE increase (2.92 $\rightarrow$ 3.42 cm) stems from a small subset of positions with large errors.

Section VII-B (page 12) now includes, after the sentence reporting the WLS degradation:

“These terms arise because all ratios share the measurement $\hat{\mu}_1$ (38): its noise perturbs every constraint normal $\hat{\mathbf{a}}_i$ coherently along $\mathbf{n}_{t,1}$ and hence does not average out with K , so GLS suppresses it through Σ_β^{-1} whereas WLS weights it as independent noise—a mismatch that grows with the number of correlated ratios. The penalty is confined to the RMSE, as the APE and $\text{CDF}_{90\%}$ of WLS still improve from $K = 5$ to $K = 9$, i.e., it affects only a small subset of positions with large errors.”

2. *The reviewer suggests explicitly defining how the baseline SNR value of 14dB used in Section VII was chosen or whether it represents a typical operational indoor environment.*

We thank the reviewer for this comment. The 14 dB operating point is the testbed-averaged SNR obtained from (6a)–(6c) with the system parameters and assumed receiver-noise variance listed in Table II over the 1,792 receiver positions of the testbed. It is therefore a reference operating point for the adopted simulation configuration, rather than a value intended to characterize a universally typical indoor environment. Since the operating SNR of a given deployment depends on the hardware and on the ambient-noise level, the performance is characterized as a function of SNR for both the bounds (Fig. 5) and the estimators (Fig. 10), and Section VII-C explains how the SNR axis can be read in terms of ambient light.

Section IV (page 7), where the 14 dB value first appears, now defines it and quotes the bounds at this operating point, consistent with Tables III and VI:

“At the nominal operating point (SNR = 14 dB, the testbed-averaged SNR (6c) corresponding to the system parameters of Table II), the $K = 5$ set achieves an RMS-DEB of 0.52° and an RMS-PEB of 1.64 cm.”

Section VII-C (page 12) refers to this definition: “[...] at the nominal operating point (SNR = 14 dB; cf. Section IV) [...]”. Fig. 5 has been re-plotted on the same absolute SNR scale as Fig. 10, so that the nominal operating point is located at 14 dB in both figures, and the accompanying sentence has been corrected accordingly.

3. *The reviewer thinks you should add a brief sentence in Section IV explaining how the initial population or search grid of the Genetic Algorithm was initialized to show it avoids getting stuck in local minima.*

We thank the reviewer for this suggestion. The initial population of $P = 300$ individuals is generated by sampling each gene (tilt or azimuth) independently and uniformly at random within its interval of the bounded search range Ω (Table II), which, for at most $2K = 18$ variables, provides a dense multi-point coverage of the search space. The population then evolves through stochastic uniform selection, scattered crossover ($p_c = 0.8$), and adaptive feasible mutation, which preserves diversity throughout the run. Section IV (page 6) now reads:

“The initial population of P individuals is generated by sampling each gene independently and uniformly at random over its admissible interval in Ω , providing a dense, unbiased multi-point coverage of the $2K$ -dimensional search space. Standard genetic operators—stochastic uniform selection, crossover with probability p_c , and adaptive feasible mutation—are applied for at most $G_{\max}$ generations, with early termination when the average relative improvement of the best fitness over $G_{\max}$ generations falls below 10^{-6} . This population-based exploration, combined with the diversity preserved by the mutation operator, makes the search robust against entrapment in local minima.”

While revising this paragraph, the selection operator, previously described as “tournament selection”, was corrected to “stochastic uniform selection”, which is the operator used in the implementation.

4. *The reviewer recommends clarify whether the proposed direction-finding algorithms can still maintain their mathematical independence from the receiver orientation if the non-line-of-sight multipath reflections become dominant.*

We thank the reviewer for this pertinent question. The independence established in Section VI-C relies on the LOS channel model (1)–(3), under which all receiver-dependent terms enter the K mean powers through the common factor $\eta = C \cos \psi / d^2$, which cancels exactly in the power ratios (29) of GLS/WLS and in the normalization (24) of NLS. Under NLOS multipath, each received power also includes reflected contributions that depend on the LED steering orientation and on $\mathbf{n}_r$; they do not share a common factor and therefore cancel in neither (29) nor (24). The independence then holds only approximately, and it is no longer guaranteed when NLOS reflections become dominant. In indoor optical channels with an unobstructed LOS, the LOS component dominates the reflected contributions [44]; the explicit modeling of NLOS propagation is left for future work. The manuscript has been modified in two places. Section VI-C (page 10) now closes with:

“Its validity under non-line-of-sight (NLOS) multipath is discussed in Section VII-D.”

and Section VII-D (page 13) contains the discussion:

“The LOS channel model also delimits the receiver-orientation independence of Section VI-C: under NLOS multipath, the received powers include reflected contributions that depend

on both the LED steering orientation and $\mathbf{n}_r$, and cancel in neither (29) nor (24), so the property holds only approximately and is not guaranteed when reflections dominate; in indoor channels with an unobstructed LOS, however, the LOS component dominates the reflected contributions [44].”

5. *It is recommended to add discussions on the applications of the proposed scheme in edge computing and digital twins. The related works can be found in reliability-aware computation offloading for delay-sensitive applications in mec-enabled aerial computing, iee transactions on green communications and networking, generative ai-aided qoe-aware resource allocations for ris-assisted digital twin interaction with uncertain evolution, iee transactions on mobile computing, digital twins for low-altitude uav networks–cooperation and learning, iee transactions on mobile computing. These studies can provide rich insights in these fields, and the authors should discuss all of them in this work.*

We thank the reviewer for pointing us to these works. Since the paper is focused on the positioning method, this discussion has been integrated in the Introduction, where the application context of indoor positioning is presented. The three works are now cited as [13], [14], and [15]. The second paragraph of Section I (page 1) now ends with:

“Beyond these use cases, such accuracy underpins emerging edge-computing and digital-twin services, from reliability-aware computation offloading [13] and quality-of-experience-aware resource allocation for digital-twin interaction [14] to cooperation and learning in low-altitude aerial networks [15].”

6. *The reviewer suggests double-checking the header of Table II and the text body, as the room dimensions are listed as 3x3x2 cubic meters but the x and y ranges are specified with a pi subscript which looks like a typo.*

We thank the reviewer for the careful reading of Table II. The room dimensions ($3 \times 3 \times 2 \text{ m}^3$) and the receiver ranges ($x, y \in [-1.5, 1.5] \text{ m}$, $z \in [0, 1.2] \text{ m}$) are consistent: the transmitter is located at the center of the ceiling, $\mathbf{t} = [0, 0, 2]^\top \text{ m}$, so the horizontal ranges span the full $3 \times 3 \text{ m}^2$ floor. The symbol that resembles a “pi subscript” is the azimuth angle φ_i of the i -th LED orientation in the row “Search range”, which specifies the intervals of the GA decision variables (tilt θ_i and azimuth φ_i , Section IV) rather than the spatial ranges of the testbed. To avoid any ambiguity, this row of Table II (page 6) now labels both angles explicitly:

“Search range (Ω): θ_i (tilt) $\in [0, 80^\circ]$, φ_i (azimuth) $\in [0, 360^\circ]$ ”